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Camera Based Parking Slot Detection For Autonomous Parking

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Abstract. Autonomous parking is one of the primary functionalities of intelligent vehicles. This requires the accurate detection of a free parking slot and its dimensions. Here we propose a method using a novel combination of computer vision techniques on the camera data mounted on sides of a car. Our occupancy check method is based on perspective guided motion segmentation, which makes it a generic approach for handling random obstacles without prior knowledge. When tested on our dataset collected in environments such as shadows, wet surface, indoors and outdoors with obstacles including cars, motorbikes, cones, carton boxes and trees, this method achieved a promising performance with F1 score higher than 97%. With the ability to run on low computational devices such as CPU, this method is adaptable to practical solutions for both AD and aftermarket ADAS systems.

Keywords: Self Parking, Motion Segmentation, Occupancy Check, Autonomous Driving (AD), Intelligent Transportation

1 Introduction

Autonomous Driving (AD) and Driver Assistance Systems (ADAS) are emerging areas to enhance human safety and driving experience. This is achieved by mounting a variety of sensors to capture data, running advanced data processing algorithms to evaluate the vehicle surroundings and making appropriate decisions for navigation. Autonomous parking is a salient functionality of AD, as a vehicle needs to find a suitable parking space while not in operation or for charging. To park a vehicle autonomously, it is required to understand the parking slot position, dimension and occupancy status. Various sensors used for this functionality include Lidars, Ultrasonic range detectors, Radars and/or cameras. These sensors are used in combination or independently to understand the parking environment. Techniques based on Lidar sensors [1] analyze point cloud data to check for the presence of obstacles and suitability of available space for parking based on the dimensions of the subject vehicle. Ultrasonic range finder sensors, associated with wheel encoder, [2], [3], [4] measure the free space information as the vehicle navigates while searching for a suitable location. These techniques can find a suitable free space for parking but have limitations in providing the

accurate markings and boundary of the parking slot. For example, these methods work decently well when a free parking slot is available next to an occupied slot or any physical object such as a pillar. These methods use the object as a reference to estimate the free slot dimension and propose an area for parking. If all the parking slots are free and there is no object for reference, these methods may propose an area which occupies more than one demarcated parking slot and the proposed orientation may not match with the designated parking slot orientation. The demarcation of a parking slot indicates the area allocated for each vehicle. Failure in detection of the accurate boundary of the parking slot may lead to an improper parking manoeuvre and end up in occupying more than one slot and/or in a wrong orientation. Using cameras as the primary sensors can help in addressing this issue, as cameras provide high-resolution information of the environment. [20] and [21] present techniques using a combination of camera and ultrasonic sensors, where cameras are used for slot marking detection and ultrasonic sensors are used for occupancy check. Fig. 1 depicts sample images of parking slots under different conditions.



Fig. 1. Sample parking slots

Several techniques are proposed which use camera data to determine a parking slot and further check for occupancy. Wang et al. [5] proposed a method using vision sensors and heuristic-based gradient analysis to determine free slots. This method uses gradient edges to find the boundaries of a slot and uses edge pixel count to decide on occupancy status. This approach works well for parking floors with uniform gradients such as concrete or asphalt, with no shadows or dirt. However, this gives poor results for slots with brick flooring, shadows, dirt or parking signs on the floor as some examples shown in Fig.1. Houben et al. [6] proposed a method using Linear Discriminant Analysis (LDA) using an image as input. This method performs well in near ideal conditions with no noise. In real-world scenarios with lots of variations in parking zones and illumination conditions, collecting the data with all variations and fine-tuning this method is a challenge. Other state-of-art methods [7] use machine learning for locating and guiding for autonomous parking. Rathour et al. [8] proposed a deep learning network which operates on front and rearview camera data and provides steering angle for end-to-end parking. Deep learning based general object detection techniques such as YOLO [9], [10] can help in detecting objects in parking slots. Jensen et al. [11], Amato et al. [12] and Acharya et al. [13] presented a specialized deep learning network for detecting free slots of a large parking area using static

cameras mounted at a high position such as the roof of a building. With very good generalization capabilities, learning-based methods show promising results in solving computer vision problems such as object detection. However, the performance of these methods depends on the size and variation of the training dataset covering all possible variations of shape, color, obstacle types and illumination conditions. Based on the complexity of network architecture, machine learning methods demand high computational resources such as GPUs (Graphics Processing Units). This may limit their application for a low-cost real-time solution. Due to these limitations of the existing methods [5],[6],[8],[9],[11],[12],[13], we define the requirements for parking slot detection algorithm as below: 1. A generic approach which can handle dynamic parking environments and random obstacles 2. Accurately detect parking slot boundary and occupancy status, at par with learning-based methods 3. Real-time performance 4. Low-cost sensor and computing system

In this work, we discuss a novel method to address these requirements. We propose to solve the problem of detecting a free slot in two steps using a camera image. The first step, detection of parking slots using gradient and structure of the parking lines; and second step to check for occupancy based on motion analysis which provides reliable occupancy status under varying conditions. Similar to the methods based on ultrasonic, this algorithm makes an assumption that the vehicle is in motion (this is required for motion flow estimation). We believe that this is a valid assumption as the vehicle navigates at low speeds in the process of searching for a suitable parking slot. The main contributions of our work include 1. A novel motion segmentation method by considering perspective homography of the camera. 2. A real-time and generic occupancy check method which does not require prior training of the obstacles. 3. Formulation for detecting the valid pair of parking slot corners based on spatial and geometric characteristics.

The organization of the paper is as follows. Section 2 briefs the camera calibration process. Section 3 deals with the detection of parking slot marks. Section 4 talks about the occupancy check method. Section 5 presents the complete system, experiments and Results followed by Conclusions in Section 6.

2 Camera Setup and Calibration

To address the problem of detecting parking slots, a camera with a wide-angle lens is mounted on one or both sides of the vehicle as shown in Fig. 2. Pitch angle and mounting position of the camera is selected in such a way that an area between 0.3 meters to 5 meters from the camera are covered in the image view. Camera calibration is performed by placing a reference planar marker on the ground at a known distance. The calibration process generates a homography matrix h , which helps in mapping image to world coordinates and vice-a-versa. For the benefit of faster performance, we generate and store reference lookup tables for image-to-world coordinate mapping and world-to-image coordinate mapping. We also compute the perspectivity ratio map (PR) at each row in the

image, which indicates the scaling effect of perspectivity on a unit length in the world system at each row of the image.

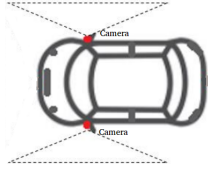


Fig. 2. Camera setup on vehicle
(Better viewed in color)

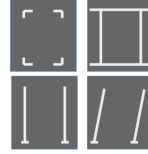


Fig. 3. Various types of parking slot markings

3 Parking Slot Marks Detection

Parking slots, in general, are represented by a set of line-markings in red, yellow or white color. These markings indicate the length, width and orientation of the parking slot allocated for each vehicle. Fig. 3 indicates a few variants of parking slot markings that are commonly used. Accurate detection of a parking slot is the first step for successful autonomous parking. The structural characteristics of parking slot markings have similarity with road lane markings in a narrow zone which can be extracted using characteristics such as edge, structure [14] and intensity thresholding [15]. Here we propose to use the gradient to locate the markings and validate by structural characteristics.

We define a pair of 1-D image filters of length $(2N + 1)$, as shown in Fig. 4. These filters help to compute rising (Fig. 4 a) and trailing (Fig. 4 b) edges in the image. Each of these filters f is applied on every pixel x in the input gray image I as shown in eq. 1, to determine the candidate rising and trailing edge pixels. A valid parking line mark is a combination of rising and trailing edge pixels separated by a distance similar to line mark thickness. So the rising and trailing edge pixels are validated by the condition that rising pixel followed by trailing pixel within a distance less than a threshold value. Then the center pixel of the line mark is computed by averaging each rising and trailing pixel position. This operation creates a binary image, where the bright pixels indicate the centerline of markings. These pixels are processed to compute approximate line segments [16], which are used for further processing. The output of applying 1-D filters and determining center pixel is shown in Fig. 5.

$$f \circ I(x) = \sum_{i=-N}^N f(i)I(x+i) \quad (1)$$

The filters shown in Fig. 4 help in determining vertical line-markings. Horizontal line-markings are determined by rotating the input image by 90 degrees

and running the same filters. The resultant line segments are rotated by -90 degrees to create horizontal line segments. Alternatively, horizontal lines can be detected by applying a set of vertical 1-D column filters. The approximate line segments are mapped to world coordinates using the homography information h created during the calibration process explained in Section II.



Fig. 4. 1-D filters a. Rising Trailing **Fig. 5.** a. Gray Image b. Edge of vertical line mark c. Edge of horizontal line mark and d. Combined vertical and horizontal edge

These line segments in world coordinates are then checked for a set of valid intersecting pairs of a vertical line and a horizontal line (V-H). This is done by shortlisting the V-H pairs with the shortest distance between vertical and horizontal line segments is less than a predefined distance threshold d_{thd} . The offset d_{thd} , which is typically equal to the thickness of the line marking, is considered to allow small gaps between V-H line segments due to pixel rounding off in edge detection and line fitting methods. For the shortlisted V-H pairs, we perform the inner product of each vertical line segment with each horizontal line segment as shown in eq. 2. Here we do not limit the angle of the V-H pair to 90 degrees to support slots that are at different orientations with respect to the horizontal boundary (as shown in the 4th sample in Fig.3).

$$\theta = \arccos(V.H/|V||H|) \times 180/\pi \quad (2)$$

Each of the valid V-H pairs is considered as a candidate parking slot corner. It is observed that some of the objects in the scene have similar intensity and geometric pattern and produce the V-H pairs. Examples of part of a vehicle forming a false V-H pair is shown in Fig. 6. To handle the false V-H pairs and to find a correct set of two V-H corners we compute a confidence score which evaluates the geometrical relationship between the set of two corners. The confidence score uses three characteristics a) Width of the parking slot should be within a predefined range b) The vertical lines in the two V-H pairs should be parallel. c) Supplementary angle property to validate the corner pair. The confidence score is calculated as shown in eq. 3a, 3b, 3c and 3d. C_w , C_v , C_{hv} and C indicate individual scores of the candidate pair of corners forming the slot based on width characteristic, vertical line angle characteristic, angle relation between the V-H pair and overall confidence score respectively. W_R and W_A indicate reference slot width and the actual distance between two corner points. ϕ_L and ϕ_R indicate the angle of vertical lines forming the left corner and right corner respectively. θ_L and θ_R represent the angle between the V-H pair of left and right corners respectively, which is calculated as shown in eq. 2.

$$C_w = 1 - (abs(W_R - W_A)/W_R) \quad (3a) \quad C_v = 1 - (abs(\phi_L - \phi_R)/90) \quad (3b)$$

$$C_{hv} = 1 - (abs((\theta_L + \theta_R) - 180)/180) \quad (3c) \quad C = C_w \times C_v \times C_{hv} \quad (3d)$$

Results of proposed slot detection method are depicted in Fig.7. Red lines indicate the nearest horizontal bound along with left and right vertical bounds of the slot. Fig. 8 depicts the flow for the proposed parking slot detection.

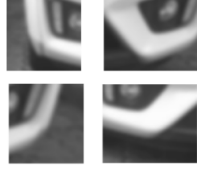


Fig. 6. Samples of image patches forming false V-H pattern



Fig. 7. Results of slot detection method

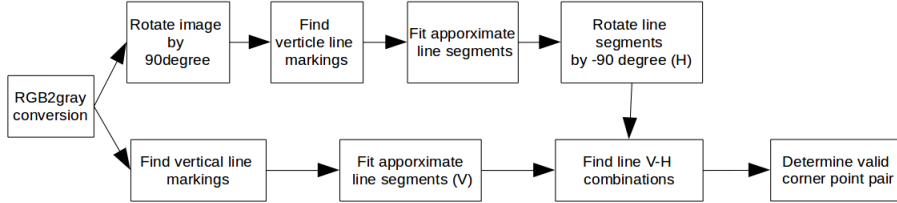


Fig. 8. Flow diagram for parking slot detection

4 Occupancy Check

After determining a parking slot region, it is processed further to check the occupancy status. Wan et al.[17] propose a moving object segmentation by motion analysis. As they consider rigid objects such as parked cars as background, such techniques are not suitable for occupancy check. We propose a novel method based on motion flow segmentation with reference to camera view to determine the presence of obstacles in the parking slot.

4.1 Motion Calculation

Optical flow is a technique to calculate the motion of all image pixels (dense flow) [18] or selected key pixels (sparse flow) [19] between two images captured with a small time delay. Optical flow operates on two constraints: a) color constancy b) small motion of corresponding pixels between two frames. Here we use the dense optical flow method [18] to produce a smooth motion field across pixels of the image. The motion flow u is computed between the current and previous frame in the parking slot detected by the process discussed in Section 3.

4.2 Homography Guided Reference Motion Estimation

To extract the obstacles, first, we compute a reference ground plane motion (ug). ug represents the motion flow of the ground plane for the current pair of frames. This is computed as explained below. Based on the current vehicle speed, we compute the displacement between current and previous frames in the world coordinate system. Then the equivalent pixel displacement at each image row is computed based on homography matrix h and reference perspectivity ratio map PR calculated as part of the calibration process.

To address the scenarios where the vehicle speed is not available, we propose an alternative method to use dense motion computed in the previous step. Mean motion at a seed row s , u_s , is computed as shown in eq 4. Typically, seed row s is a row indicating the ground plane. We use slot marking pixel value as a reference to select s . Then, motion at each image row j of width N pixels is computed using u_s and perspectivity ratio map (PR) as shown in eq 5.

$$u_s = 1/(N) \times \sum_{i=0}^{N-1} u(i) \quad (4) \quad ug_j = u_s \times PR_j / PR_s \quad (5)$$

4.3 Obstacle Detection using Homography Guided Motion Segmentation (HGMS)

The reference motion is compensated by subtracting ug from dense flow map u to compute the obstacle motion uo . To handle any noise in the flow computation, we further apply a homography guided adaptive threshold A_j to produce binary masks of potential obstacle regions as shown in eq. 6a, 6b and 6c. Here T is the threshold for a minimum motion for the seed row s . A_j is defined with perspectivity in view and gradually increasing with row number. These binary masks are further post-processed with binary morphology closing operation to eliminate breaks due to the smooth texture of any obstacle. Finally, a filter is applied based on the pixel area to suppress noise blobs of small size and retaining masks of the obstacles in the selected parking region in the current frame.

$$uo_{i,j} = u_{i,j} - ug_j \quad (6a) \quad A_j = T \times PR_j \quad (6b)$$

$$ObstacleMap_{i,j} = \begin{cases} 1, & \text{if } uo_{i,j} \geq A_j \\ 0, & \text{otherwise} \end{cases} \quad (6c)$$

Fig. 9 represents output of each step followed for obstacle detection using HGMS. Here dense motion u (Fig. 9c) is computed from two frames shown in Fig. 9a and Fig. 9b. Computed reference ground motion ug is shown in Fig. 9d. After subtracting the reference motion and applying threshold, the final obstacle masks are shown in 9f. It can be observed from this example that the proposed algorithm is able to detect both car and tree present in the input image. Flow diagram of the occupancy check step is as shown in Fig. 10.

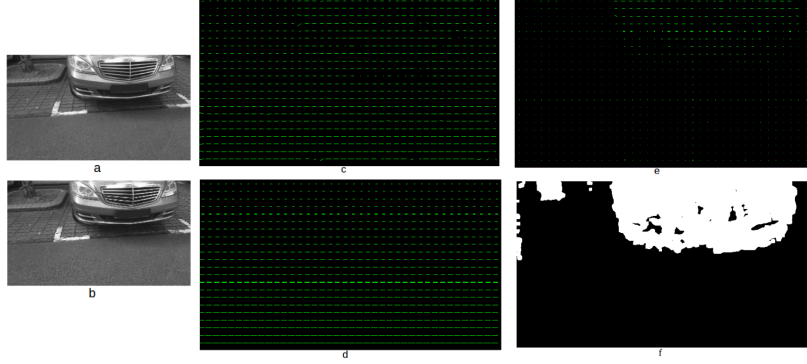


Fig. 9. HGMS a. Current frame b. Previous frame c. Motion detection using dense optical flow d. Homography guided reference motion e. Motion after reference compensation f. obstacle masks (better viewed in color)

5 Results

The well-articulated problem of free parking slot detection is addressed by the proposed algorithm. The complete system of the algorithm has been depicted as a flow diagram in Fig.11. As per our knowledge, there is no standard public dataset available for the problem of interest. So we have collected a dataset of 623 different parking slots, across different locations and at different times of the day. In the dataset, we have 402 free slots and 221 occupied slots. Among these, 65% are from the outdoors and around 10% of the slots have wet floors. Each slot data is a set of two successive frames recorded at 20fps and containing both markings of a parking slot. The comparison tests are performed on a computer

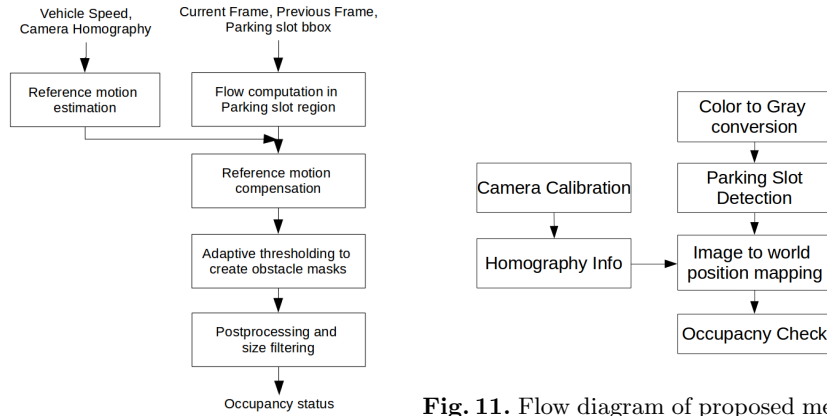


Fig. 11. Flow diagram of proposed method

Fig. 10. Flow diagram for occupancy check using HGMS

with a CPU configuration of Intel Core i7-7700HQ CPU @ 2.80GHz, 8 core processor.

The performance of HGMS method has been compared with different state-of-art algorithms such as gradient intensity based occupancy check and deep learning based object detection. Wang et al. [5] proposed to compare the pixel gradient in the region of parking slots to check for occupancy. We have compared the results of a similar algorithm which checks for the total number of pixels with a gradient greater than a threshold value. Though the gradient method requires low computational power, the challenge is to determine the optimal threshold for gradient threshold and pixel count to decide as occupied. For this comparison, we have experimentally determined these thresholds which work for the majority of the scenarios in our dataset. It is obvious that these values can not be generalized for real-world variations in the parking slot data.

As several researchers highlight the advantage of deep learning in free slot detection using static cameras, we have considered a generic object detection algorithm [9] as part of our comparison. YOLO (You Only Look Once) is a state-of-art network for generic object detection which has support for several objects including vehicles, motorcycles, bicycles and pedestrians. When we used off-the-shelf models, the performance is poor and this is due to the unusual view angle of the parking camera and including objects such as construction cones. So we custom trained the YOLO network using transfer learning and included a set of 400 images for our problem. These images are not part of our test dataset. While inference, we use an input resolution of 160x160 which takes around 120msec per frame.

For evaluating single-frame based methods such as gradient intensity and deep learning, we use the first frame of each parking slot frame sequence. Whereas for the proposed motion-based method, we use two successive frames of the dataset of each slot. Table I shows the comparison of different occupancy check

methods for a set of real parking slot images shown in the first column. We considered practical scenarios including vehicles of different colors (rows 1 and 2), shadows (row 3) and obstacles such as traffic cones (row 4). The masks of obstacles detected using a gradient intensity based method similar to [5] are depicted in the second column. The third column shows object bounding boxes in blue rectangles detected using general object detection [9] and the last column presents masks of obstacles computed by the proposed method. The gradient-based method could detect the vehicles but failed to detect an empty slot. Shadows and floor gradient, affect this method as it is entirely based on gradient variations. And gradient intensity methods fail to detect obstacles other than vehicles. The default deep learning model fails to detect unknown objects such as cones but after custom training, the deep learning model detects well for different objects and works fine with shadows. But support for any new type of obstacle requires data collection, annotation and custom training.

Results from the proposed method show that it can handle the challenging scenarios of parking slots very well. This method is not sensitive to the color or shape of the obstacles and shows superior occupancy detection for different textures and shadows in the parking slots. It is able to detect objects of different sizes such as cars, trees, traffic cones etc. This is primarily due to the underlying principles of motion estimation which are immune to these variations. With an F1 score of 97%, the performance of our method is at par with the deep learning method in handling known obstacles, shadows, dirt, wet surfaces. In addition, our method does not require any prior knowledge of the obstacles and can handle any kind of obstacles. HGMS takes around 45msec time to process a frame on a test computer with near real-time performance.

The proposed method performs well in a wide range of real-world scenarios with good to moderate quality. It has shown poor performance in detecting slot markings when occluded by other objects or low illumination conditions. These are mainly caused due to the limitations of primary sensors, i.e cameras, used for the application.

To summarize our experiments, we evaluated these algorithms to detect free parking slots. Table II shows the performance of different methods. The performance indicators Precision (P), Recall (R) and F1 Score ($F1$) are computed. From the well designed comparative analysis using a real dataset, it is observed that the gradient method is simple and fast but has limitations in handling scenes with varying intensities. Custom deep learning method shows robust behaviour in varying scene intensities with known object types. But the performance is poor when unknown obstacles are present in the scene and slowest among the methods compared. The learning-based methods are also highly dependent on the quality of and variations in the dataset used for training. Proposed motion-based method handles the problems of intensity variations, real-time and has robust performance for any type of obstacles without prior knowledge.

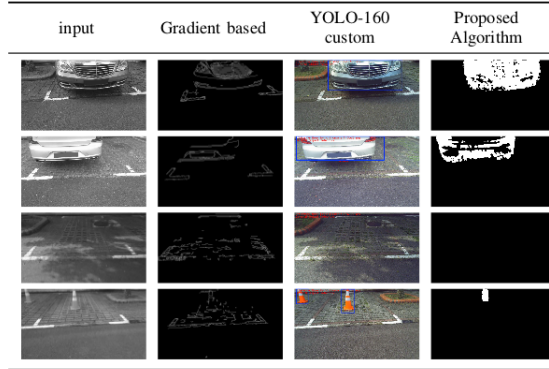


TABLE I
COMPARISON OF GRADIENT THRESHOLD, YOLO-160 AND PROPOSED
METHODS

Method	Precision	Recall	F1 Score
Gradient	0.91	0.75	0.82
YOLO-160 default	0.81	1.0	0.89
YOLO-160 custom	0.96	1.0	0.98
HGMS (proposed)	0.98	0.95	0.97

TABLE II
PRECISION AND RECALL FOR GRADIENT
THRESHOLD, YOLO-160 AND PROPOSED METHOD

6 Conclusions

Safe parking under challenging scenarios is a crucial requirement for an autonomous parking solution. The proposed free parking slot detection method shows commendable results under various real-world scenarios and stands out to be a generic solution. This method detects and validates candidate parking slot markers to eliminate noise or false patterns. Usage of the motion flow makes the proposed method insensitive to shadows and illumination variations and can detect obstacles of different sizes such as cones, trees or any other obstacle without any prior knowledge. As this method is developed based on computer vision techniques, it requires low computational power and is suitable for a low-cost parking solution. This makes our algorithm the right choice for unknown environments and suitable for both AD and ADAS. Currently, we are working on determining free space dimensions from the motion map to handle scenarios with partial or no visible parking lanes. The said work will be reported in near future.

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